HW2_search

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CS 331 **Introduction to Artificial Intelligence** Fall 2025

OREGON STATE UNIVERSITY

School of Electrical Engineering and Computer Science

Instructor: Sandhya Saisubramanian

Total points: 100 HW 2: **Heuristics, Local Search and Adversarial Search** Due date: Oct 24, 2025

Instructions: This homework assignment consists of a written portion and a programming portion. Collaboration is not allowed on any part of this assignment. Solutions must be typed (hand written and scanned submissions will not be accepted) and saved as a .pdf file. You will submit a single .zip file that contains the code base and solutions as a .pdf file.

1. **(10 points)** Consider the delivery truck problem discussed in Lecture 2, with s1 as the start node and s16 as the goal node (Figure 1). Calculate the heuristic value for each node in this problem, by computing the Manhattan distance between a node n and the goal node. You must calculate the heuristic value of all nodes, including the start node, goal node and the nodes marked as construction areas (denoted by shaded cells). For reference, the xy coordinates of $s1 = (0, 0)$ and the xy coordinates of $s13 = (0, 3)$. The Manhattan distance between two nodes denoted by (x_1, y_1) and (x_2, y_2) is calculated as $|x_1 - x_2| + |y_1 - y_2|$. For example, the distance between $(1, 2)$ and $(4, 6)$ is $|1 - 4| + |2 - 6| = 3 + 4 = 7$.

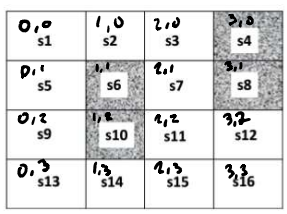

Figure 1: Delivery truck problem

1.

s1: 3 + 3 = 6
s2: |1-3|+|0-3| = 5
s3: |2-3|+|0-3|=4
s4: |3-3|+|0-3|=3
s5: |0-3|+|1-3|=5
s6: |1-3|+|1-3|=4
s7: |2-3|+|1-3|=3
s8: |3-3|+|1-3|=2
s9: |0-3|+|2-3|=4
s10: |1-3|+|2-3|=3
s11: |2-3|+|2-3|=2
s12: |3-3|+|2-3|=1
s13: |0-3|+|3-3|=3
s14: |1-3|+|3-3|=2
s15: |2-3|+|3-3|=1
s16: 0

2. **(20 points)** We will now apply Local Beam search for a goal-oriented problem from HW1, even though it isn't designed for this. The purpose of doing this is to compare the performance of different search algorithms on the same problem.

   Solve the graph in Fig. 2 using Local Beam Search with $k = 2$ (i.e. always maintain two "best" nodes and generate all their successors). The evaluation function to identify the "best" nodes in the current beam (open list) is g (the accumulated cost from the start node). The search begins by expanding the successors of the start node. The stopping criteria for the search is: stop when the **goal node is generated**. Answer the following questions: (i) was Beam search able to find a solution?; (ii) is it an optimal solution? You can either solve this question manually by hand or code it up. If you are coding it up, include a screenshot of the output in the writeup and include your code in the submission. Clearly explain how you break ties.

2.

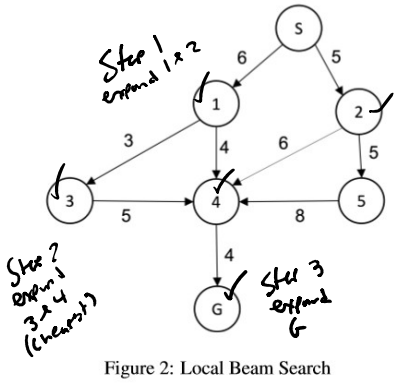

Figure 2: Local Beam Search

i. The solution:

S1 4 G

ii. It is not an optimal solution like A* since it only considers local heuristics unlike A* and could have thrown away a node that would lead to the optimal path if it wasn't one of the two cheapest.

3. **(30 points)** Simple LLM decoding using Beam search. Assume you are manually decoding from a simple language model with the vocabulary: $\{\text{I}, \text{like}, \text{love}, \text{cats}, \text{dogs}, \text{and}, \langle\text{eos}\rangle\}$.

The conditional probabilities are as follows:

- From $\langle s\rangle$: $P(I) = 0.7, P(\text{cats}) = 0.1, P(\text{dogs}) = 0.1, P(\text{like}) = 0.1$
- Given the most recent token is "I": $P(\text{like}) = 0.6, P(\text{love}) = 0.3, P(\langle\text{eos}\rangle) = 0.1$
- Given the most recent token is "like": $P(\text{cats}) = 0.5, P(\text{dogs}) = 0.4, P(\langle\text{eos}\rangle) = 0.1$
- Given the most recent token is "love": $P(\text{cats}) = 0.6, P(\text{dogs}) = 0.2, P(\langle\text{eos}\rangle) = 0.2$
- Given the most recent token is "cats": $P(\text{and}) = 0.2, P(\langle\text{eos}\rangle) = 0.8$
- Given the most recent token is "dogs": $P(\text{like}) = 0.3, P(\langle\text{eos}\rangle) = 0.7$
- Given the most recent token is "and": $P(\text{cats}) = 0.5, P(\text{dogs}) = 0.5$

We will consider a "dummy" start state $\langle s\rangle$ that can be augmented with any word (or token) from the vocabulary. Terminal sequences are those that end with $\langle\text{eos}\rangle$. Use Beam search with $k = 2$ (maintains 2 best sentence sequences for expansion). The evaluation function for selecting the best sequence is **cumulative probability**. The cumulative probability of a sentence is calculated as the product of the conditional probabilities of all tokens in the partial sentence. For example, the cumulative probability of the partial sentence sequence "$\langle s\rangle$ Dogs like cats" = $0.1 \times 0.3 \times 0.5$.

Clearly show the trace of Beam Search and report the final two best sentences, along with their cumulative probabilities.

3.

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Beam Search Results (k=2):

1. Sentence: "<s> I like cats <eos>"
   Probability: 0.1680

2. Sentence: "<s> I like dogs <eos>"
   Probability: 0.1176
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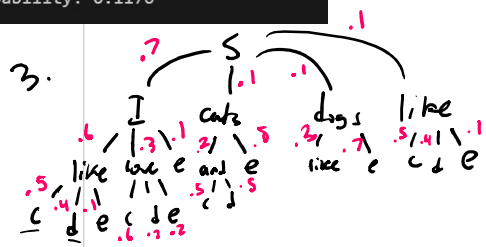
3.



4. $P = e^{(-\Delta E)/T}$

$= 2.718^{(-5)/10} = 0.60656$ move 1

$= 2.718^{(-10)/10} = 0.3679$ move 2

4. **(10 points)** Consider the problem of minimizing energy E using simulated annealing. There are two possible moves and their corresponding energy differences are: $\Delta E_1 = 5, \Delta E_2 = 10$. The temperature parameter $T = 10$. Calculate the probability with which each of these moves will be executed. Which move is more likely? Use $e = 2.718$ for the calculation.

5. **(30 points)** Consider the graph in Figure 3 with blue nodes denoting MAX nodes and red nodes denoting MIN nodes.

(i) Solve the graph using minimax algorithm. Clearly show your calculations at each step. Identify what strategy A must choose and the corresponding payoff it will receive (value).

(ii) Solve the graph using alpha-beta pruning. Clearly show your calculations at each step. What strategy should A choose and what is the corresponding payoff?

(iii) Which nodes are pruned using alpha-beta pruning? Did it change A's strategy? Why or why not?

i.

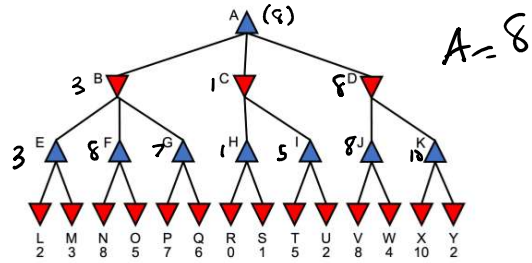
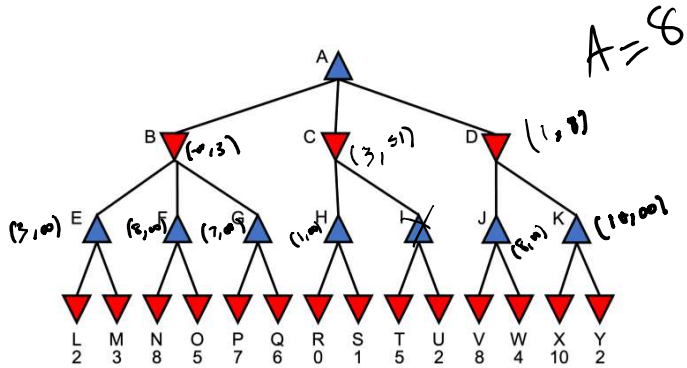


Figure 3: Minimax graph

ii



iii. We pruned node I, it didn't really affect A's strategy as A's choice would not end up being in that subtree as it is the smallest min it could have selected of 3, 1, or 8.

i.

1. E chooses M(3)
2. F chooses N(8)
3. G chooses P(7)
4. B chooses E(3)
5. H chooses S(1)
6. I chooses T(5)
7. C chooses H(1)
8. J chooses V(8)
9. K chooses X(10)
10. D chooses J(8)
11. A chooses J(8)

ii.

1. E max(2,3) = 3, B min(inf, 3)
2. F max(3,8) = 8, B min(3,8)
3. G max(3,7)= 7, B min(3,7)
4. B chooses E(3), A max(-inf, 3)
5. H max(0,1)=1, C min(inf, 1)
6. Prune i since 3 > 1
7. J max(8,4)=8, D min(inf, 8)
8. K max(10, 2)=10, D min(10,8)=8
9. A chooses 8